PRACTICE SHEET

1. What is the product of first 2n + 1 terms of a geometric progression?
(a)The (n + 1)th power of the nth term of the GP
(b)The (2n + 1)th power of the nth term of the GP
(c)The (2n+1) the power of the (n+1)th term of the GP
(d)The nth power of the (n + 1)th terms of the GP

2. If x + 1, 4x + 1, and 8x + 1 are in geometric progression, then what is the non – trivial value of x?
(a) – 1 (b) 1
(c) 1/8 (d) 1/4

3. The equation $(a^2 + b^2)\,x^2 - 2b\,(a + c)\,x + (b^2 + c^2) = 0$ has equal roots. Which one of the following is correct about a, b, and c?
(a) They are in AP
(b) They are in GP
(c) They are in HP
(d) They are neither in AP, nor in GP, nor in HP

4. For an AP with first term u and common difference v, the pth term is 15 uv more than the qth term. Which one of the following is correct?
(a) p = q + 15v (b) p = q + 15 u
(c) p = q + 14 v (d) p = q + 14 u

5. If a, b and c are three positive numbers in an arithmetic progression, then :
(a) $ac > b^2$ (b) $b^2 > a + c$

12. Sum of first n natural numbers is given by $\frac{n(n+1)}{2}$. What is the geometric mean of the series 1, 2, 4, 8, ... 2^n?
(a) 2^n (b) $2^{n/2}$
(c) $2^{1/2}$ (d) 2^{n-1}

13. If the number of terms of an A.P. is $(2n + 1)$, then what is the ratio of the sum of the odd terms to the sum of even terms?
(a) $\frac{n}{n+1}$ (b) $\frac{n^2}{n+1}$
(c) $\frac{n+1}{n}$ (d) $\frac{n+1}{2n}$

14. If he sum of 'n' terms of an arithmetic progression is $n^2 - 2n$, then what is the nth term?
(a) $3n - n^2$ (b) $2n - 3$
(c) 2n + 3 (d) 2n – 5

15. If a, 2a + 2, 3a + 3 are in GP, then what is the fourth term of the GP?
(a) –13.5 (b) 13.5
(c) –27 (d) 27

16. What is sum to the 100 terms f the series 9 + 99 + 999 + ...?
(a) $\frac{10}{9}(10^{100} - 1) - 100$